Anaphora Resolution -- Assignment Report

Problem 1 (10 Marks): Extracting Candidate Noun Phrases using a POS Tagger

Methodology: Given a discourse segment and a target pronoun at character offset c_p, we identify all candidate Noun Phrases (NPs) using NLTK's Part-of-Speech tagger (nltk.pos_tag_sents) and a Context-Free Grammar (CFG) chunk parser (nltk.RegexpParser). Each segment is split into sentences (nltk.sent_tokenize) and words (nltk.word_tokenize) while tracking exact character offsets (start_char, end_char). We parse base NPs ({<DT>?<JJ.*>*<NNP.*>+}, {<DT>?<JJ.*>*<NN|NNS>+}) and recursive possessive/compound NPs ({<NP><POS><NP>}, {<NP><NP>}), extracting all non-pronominal NPs along with their POS tags, spans, and grammatical roles.

* Total test discourse segments (gap-test.tsv): 2,000
* Total candidate NPs extracted: 42,597 (Average: 21.30 candidate NPs per segment)
* Sample Segment (test-1, pronoun 'His' at offset 383): 26 candidate NPs extracted, including 'the Kontinental Hockey League', 'Dehner', 'a contract', 'Former NHLer Gary Suter', 'Olympic-medalist Bob Suter', 'uncles'.

Problem 2 (5 Marks): Random Assignment Baseline (100 Runs)

Methodology: For each test segment in gap-test.tsv, we randomly select one candidate NP using random.choice across 100 independent runs and average the accuracy. Because GAP provides ground-truth coreference labels for two specific named candidate entities (A and B), we report both Full Candidate NP Pool accuracy (sampling across all ~21 POS-extracted NPs) and GAP Candidate Pair accuracy (sampling between A and B).

* Average Accuracy (Full POS-Extracted Candidate NP Pool, 100 runs): 7.13%
* Average Accuracy (GAP Candidate Entities A vs. B, 100 runs): 44.07% (under 50% because ~10% of segments have neither A nor B as gold antecedent)

Problem 3 (5 Marks): Most Recent Antecedent Baseline

Methodology: For each target pronoun in gap-test.tsv, we deterministically select the closest candidate NP immediately preceding the pronoun (preceding[-1]).

* Most Recent Accuracy (Full POS-Extracted Candidate NP Pool): 12.85% (257 / 2000 segments)
* Most Recent Accuracy (GAP Candidate Entities A vs. B): 40.15% (803 / 2000 segments)

Analysis: Across all extracted NPs, selecting the most recent NP nearly doubles accuracy over random guessing (12.85% vs. 7.13%), confirming that surface recency is a helpful signal. However, between A and B, surface recency (40.15%) is lower than random (44.07%) because in English clauses (Subject [A] verb Object [B], and he...), the earlier grammatical Subject (A) is more often the true antecedent than the more recent Object (B).

Problem 4 (10 Marks): Custom Logistic Regression Classifier

Implementation From Scratch: We implemented SimpleLogisticRegression in pure Python using math.exp. Given a 12-dimensional feature vector x for a (pronoun, candidate_NP) pair, the model predicts P(y=1|x) = 1 / (1 + exp(-(w^T x + b))). We trained the weights via Stochastic Gradient Descent (SGD) on gap-development.tsv (2,000 segments) with L2 regularization and positive class weighting (pos_weight) to account for the ~1:20 positive-to-negative candidate ratio. Hyperparameters tuned on gap-validation.tsv (454 segments) selected learning_rate=0.1, l2_reg=0.001, pos_weight=4.0.

* Full POS-Extracted Candidate NP Pool (gap-test.tsv): Precision = 40.58% | Recall = 45.57% | F1-Score = 42.93%
* GAP Candidate Entities A vs. B (gap-test.tsv): Precision = 56.65% | Recall = 58.60% | F1-Score = 57.61%

Problem 5 (5 Marks): Top-5 Learned Features and Linguistic Explanation

* 1. entity_frequency (weight = +4.5121): Strongest positive predictor. Entities mentioned repeatedly across the passage are the global discourse topic (Centering Theory salience).

* 2. is_proper_noun (weight = +2.5635): 2nd strongest predictor. GAP target pronouns (he/she/him/her/his) refer to animate humans; proper nouns (NNP) filter out inanimate common NPs.

* 3. grammatical_role_subj (weight = +1.0527): 3rd strongest predictor. Reflects the Grammatical Role Hierarchy (Subject > Object > Oblique); syntactic subjects are preferred antecedents.

* 4. gender_mismatch (weight = -0.9855): Strong negative weight penalizing candidate NPs whose lexical/gazetteer gender conflicts with the pronoun's gender.

* 5. is_hobbs_top1 (weight = +0.6018): Positive weight rewarding the #1 candidate favored by Hobbs' left-to-right breadth-first CFG tree search.

Problem 6 (5 Marks): Error Analysis & How the Algorithm Can Be Improved

* Error 1 -- Kinship Semantics & Possessive Modifiers (test-1): Predicted 'Former NHLer Gary Suter' (p=0.530) vs. Gold 'Dehner' for 'His'. Gary Suter is the syntactic subject, whereas resolving 'His cousin' to Dehner requires kinship world knowledge.

* Error 2 -- Quoted Speech vs. Main Narrative Frame (test-4): Predicted 'Marchesano' (p=0.823) vs. Gold 'Pisciotta' for 'his'. Marchesano is repeated inside a quote spoken by Pisciotta right before the pronoun, inflating local frequency and recency.

* Error 3 -- Character vs. Actor Parentheticals (test-5, test-6): Predicted 'William McNamara' (p=0.575) vs. Gold 'Eddie' for 'his'. In 'Eddie Devane (William McNamara) is a young sailor on his way...', the parenthetical actor name is closer to the pronoun and repeated earlier in the cast list.

* Error 4 -- Event Semantics & Selectional Restrictions (test-8): Predicted 'Sophie' (p=0.636) vs. Gold 'Jeni' for 'her'. Both are feminine subjects, and knowing that 'Jeni is killed when the monster... pulls her in' requires causal verb semantics.

How the Algorithm Can Be Improved: (1) Replace shallow regex chunking with full dependency parsing to distinguish nsubj, dobj, pobj, and poss roles; (2) Track quotation boundaries and speaker attribution so mentions inside quotes do not override the narrative speaker; (3) Detect parenthetical Character (Actor) appositives; and (4) Group partial name mentions into entity clusters and incorporate verb selectional preferences.

Problem 7 (10 Marks): Rule-Based Hobbs' Algorithm on CFG Parse Trees

Methodology: We converted each POS-tagged sentence into a doubly-linked Context-Free Grammar (CFG) parse tree (ParentedTree) and implemented the 9-step Hobbs' tree traversal (Jurafsky & Martin, Sec. 21.6.1): climbing from the pronoun NP to enclosing NP/S nodes, searching left branches in left-to-right Breadth-First order while enforcing Binding Condition B in Step 3 (requiring an intervening NP/S node), and searching preceding sentences s_{t-1}, s_{t-2}, ... left-to-right breadth-first with gender/number/person compatibility.

* Full POS-Extracted Candidate NP Pool (gap-test.tsv): Precision = 30.42% | Recall = 34.29% | F1-Score = 32.24%
* GAP Candidate Entities A vs. B (gap-test.tsv): Precision = 43.46% | Recall = 42.36% | F1-Score = 42.90%

Problem 8 (5 Marks): Comparison of Logistic Regression and Hobbs' Algorithm

Summary Table on gap-test.tsv (2,000 segments):

* Random Assignment (100 runs) : All-NPs F1 = 7.13% | GAP (A/B) F1 = 44.07% (P=44.07%, R=44.07%)
* Most Recent Antecedent : All-NPs F1 = 12.85% | GAP (A/B) F1 = 40.15% (P=40.15%, R=40.15%)
* Rule-Based Hobbs' Algorithm : All-NPs F1 = 32.24% | GAP (A/B) F1 = 42.90% (P=43.46%, R=42.36%)
* Custom Logistic Regression (Ours) : All-NPs F1 = 42.93% | GAP (A/B) F1 = 57.61% (P=56.65%, R=58.60%)

* Extra Credit (Scikit-Learn LR) : All-NPs F1 = 42.33% | GAP (A/B) F1 = 57.36% (P=55.61%, R=59.22%)

Qualitative Comparison: Hobbs' algorithm improves Full-Pool F1 from 12.85% (Most Recent) to 32.24% because left-to-right breadth-first traversal of CFG trees prioritizes syntactic Subjects over embedded prepositional objects and enforces Binding Condition B. However, Custom Logistic Regression substantially outperforms Hobbs' algorithm (+10.69% Full-Pool F1 and +14.71% GAP A/B F1) because Hobbs is memoryless across sentences--picking the first compatible subject in s_{t-1} even if it is a minor character--whereas Logistic Regression combines global discourse salience (entity_frequency), animacy (is_proper_noun), grammatical role, gender agreement, and Hobbs' syntactic ranks into a calibrated probabilistic score.

Extra Credit (10 Marks): Scikit-Learn Logistic Regression Comparison

Using scikit-learn's LogisticRegression (C=0.1, pos_weight=4.0 tuned on dev set) on the same 12 features yields 42.33% F1 on the Full Candidate NP Pool (P=39.99%, R=44.95%) and 57.36% F1 on GAP A/B (P=55.61%, R=59.22%), with the same top features: entity_frequency (+4.2958), is_proper_noun (+2.9008), grammatical_role_subj (+1.1571), recency_inv_rank (+0.9065), and gender_mismatch (-0.7172). It makes 971 errors on gold-annotated test segments with the same error patterns and outperforms Hobbs' algorithm by +10.09% Full-Pool F1 and +14.46% GAP A/B F1.